

§7.4 - Integration of Rational Functions by Partial Fractions

Definition #1 (Polynomial)

Let $a_0, a_1, \dots, a_n \in \mathbf{R}$ where $a_n \neq 0$.

Then

$$P(x) = a_0 + a_1x + \dots + a_nx^n$$

is a **polynomial** of degree n .

If $n = 1$ then P is a linear polynomial.

Definition #2 (Reducible Polynomial)

Let P be a polynomial of degree n .

If u and v are polynomials of degree less than n such that

$$P(x) = u(x) \cdot v(x)$$

then P is a **reducible polynomial**. Otherwise, P is an **irreducible polynomial**.

Definition #3 (Roots)

Let P be a polynomial and $x_0 \in \mathbf{R}$.

If $P(x_0) = 0$ then there exists a $k \in \mathbf{N}$ and a polynomial Q such that

$$P(x) = (x - x_0)^k Q(x), \quad Q(x_0) \neq 0.$$

We say x_0 is a **root** of P with **order** k .

If $k = 1$ then x_0 is a **simple root** and if $k > 1$ we have a **repeated root**.

Definition #4 (Rational Function)

Let P and Q be polynomials.

Then R given by

$$R(x) = \frac{P(x)}{Q(x)}$$

is a **rational** function.

If the degree of Q is greater than the degree of P then R is **proper rational function**.

Definition #5 (Partial Fraction Decomposition)

Let R be a proper rational function.

Then the **partial fraction decomposition** of R is

$$R(x) = \sum_{k=1}^m \frac{P_k(x)}{Q_k(x)}$$

where P_k is a polynomial and Q_k is a power of an irreducible polynomial.

(Here the degree of Q_k is greater than the degree of P_k for all k .)

Example #1 (Simple roots)

Find the partial fraction decomposition of

$$\frac{x - 3}{(x - 2)(x + 4)}.$$

Example #2

Find

$$\int \frac{x - 3}{(x - 2)(x + 4)} dx.$$

Example #3 (Repeated roots)

Find the partial fraction decomposition of

$$\frac{x - 4}{(x - 2)(x + 4)^2}.$$

Example #4

Find

$$\int \frac{x - 4}{(x - 2)(x + 4)^2} dx.$$

Example #5 (Simple irreducible factor)

Find the partial fraction decomposition of

$$\frac{3x - 1}{(x - 1)(x^2 + 2x + 5)}.$$

Example #6

Find

$$\int \frac{-x + 9}{x^2 + 2x + 5} dx$$

Example #7 (Improper rational function)

Divide

$$2x^2 + 5x + 6$$

by

$$x^2 + 3x + 2.$$

Example #8 (Repeated irreducible factor)

Find the partial fraction decomposition of

$$\frac{1}{(x+1)(x^2+2x+2)^2}.$$

Example #9

Find

$$\int \frac{e^{3x} + e^{2x} + e^x}{(e^{2x} + 1)^2} dx.$$

Example #10

Find

$$\int \frac{1}{x\sqrt{1+2x}} dx.$$